

Fig. 5: An actual-size, single-side PCB for the wireless steering

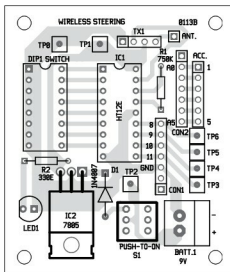


Fig. 6: Component layout for the PCB in Fig. 5

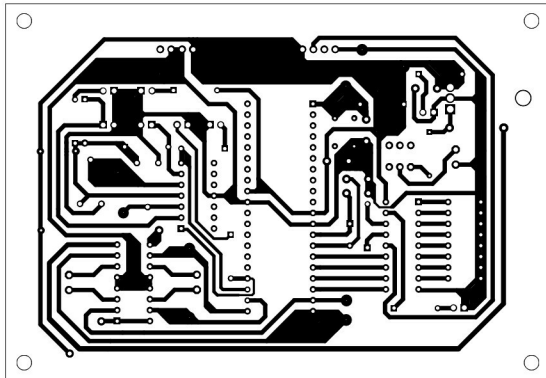


Fig. 7: An actual-size, single-side PCB for the robot

crease above the predefined value, the transmitter transmits the code for left movement to the robot. This code is received by the receiver of the robot and the microcontroller drives the motors to turn the robot to the left. Similarly,

to move the robot in another direction,
tilt the steering in that direction.

Software

The wireless steering is built around Arduino Uno board. Arduino Uno

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

is programmed using Arduino IDE software. ATmega328 on Arduino Uno comes pre-burnt with a boot loader that allows you to upload new code to it without the use of an external hardware programmer. It communicates using the original STK500 protocol. You can also bypass the boot loader and program the microcontroller through ICSP (in-circuit serial programming) header, but using boot loader programming is quick and easy. Select Arduino Uno from 'Tools→Board' menu (according to the microcontroller on your board) in Arduino IDE and burn the program through standard USB port in the computer.

The source program for the robot is written in 'C' language and compiled using MPLAB IDE for the robot. Burn the hex code into the microcontroller using a suitable programmer. Program execution starts by initialising the input and output ports. The program checks the received data for forward, backward, left and right movements and gives corresponding control signals to the motor driver.

Construction and testing

Actual-size, single-side PCBs for the wireless steering and the robot are shown in Figs 5 and 7, respectively. Corresponding component-side layouts are shown in Figs 6 and 8, re-



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